GAME HUB

Final Project Document — Code Structure and Logic

CSE Structured Programming (C99) · Multi-file console + graphics

Progress milestone: login, shell, Candy Crush

1. Project idea

GAME HUB is a terminal environment in C99. The user logs in, sees a coloured banner, and types shell-style commands. Games open from the same prompt. This progress build implements Candy fully; golf, snake and ludo are planned. Design uses structs, files, macros, multi-file layout, and **make project**.

2. Directory layout

```
gamehub-progress/
  Makefile                make project / make run / make clean
  README.txt
  build.bat
  include/hub.h           types, macros, declarations
  src/main.c              shell loop, command dispatch
  src/util.c              login, banner, files, settings stub
  src/g_candy.c           Candy Crush (graphics + terminal)
  data/users.txt
  data/ratings.txt
  data/settings.txt
```

3. Course topics used

Topic	Where
Structs	User, Settings, Session in hub.h
Pointers	Session *s, FILE *, char *cmd
2D arrays	Candy board g[rows][cols]
Text files	users.txt, ratings.txt, settings.txt
Macros	ANSI_*, SW, SH, paths, CLEAR_CMD
Header files	hub.h included by every .c file
Multi-file	main.c, util.c, g_candy.c + Makefile
Loops / if	Shell loop, match scan, gravity
Makefile	make project builds the program

4. Data structures (hub.h)

User: name and password. **Settings**: display mode, difficulty, theme, candy options. **Session**: groups user + settings so one pointer is passed into the shell and games.

```
typedef struct { char name[MAX_NAME]; char pass[MAX_PASS]; } User;
typedef struct {
    int display_mode;  /* 0 graphics, 1 terminal */
    int difficulty;    /* 1 easy, 2 normal, 3 hard */
    int theme, two_player;
    int candy_colors, candy_rows, candy_cols, candy_moves;
} Settings;
typedef struct { User user; Settings settings; } Session;
```

5. main.c — shell logic

main enables console colours, seeds rand, zeroes Session, loads settings, then login_or_register. On success print_banner runs. The infinite loop prints user@hub>, reads a line with fgets, strips newline, skips spaces, and

dispatches with strcmp.

```
Session s;
load_settings(&s.settings);
if (!login_or_register(&s.user)) return 1;
print_banner(&s);
for (;;) {
    fgets(line, ...);
    if "help"     -> print_help();
    if "ls"       -> print_ls();
    if "settings" -> open_settings_menu();  /* stub */
    if "status"   -> show_status();
    if "candy"    -> game_candy(&s); then rating;
    if "exit"     -> save_settings; break;
}
```

6. util.c — login, files, banner

login_or_register: option 1 login / 2 register. fgets for name and password; trim CR/LF. Register scans users.txt for duplicate names then appends. Login scans for matching pair and copies into User with strncpy.

load_settings / save_settings: defaults first, then fscanf/fprintf a line of integers in settings.txt.

ask_and_save_rating: after candy, asks 1–5, appends to ratings.txt, shows average_rating().

print_banner: clear screen, ASCII GAME logo, user, time, game list, ANSI colours.

open_settings_menu: progress stub — prints planned options only; does not edit values yet.

7. g_candy.c — game logic

If display_mode is 1, terminal board with ANSI backgrounds. Else raylib window.

Graphics path

Random 2D colour grid. Click selects a cell; second click on an adjacent cell swaps. Horizontal and vertical scans mark three-in-a-row as -1. No match → swap back. Match → score up, moves down, gravity packs columns, top cells refilled randomly. ESC closes window back to shell.

```
/* horizontal match */
if g[rr][cc]==g[rr][cc+1]==g[rr][cc+2] -> mark -1;
/* gravity: pack live cells to bottom of each column, fill top */
```

8. Makefile — make project

Target **project** builds the executable. On Windows it uses raylib under include/raylib-6.0_win64_mingw-w64/. **make run** builds then starts the program. **make clean** removes the binary.

```
project: $(TARGET)
■@echo Build complete: $(TARGET)

$(TARGET): src/main.c src/util.c src/g_candy.c include/hub.h
■gcc -Wall -Wextra -std=c99 -Iinclude ... -o gamehub.exe ... -lraylib ...
```

9. Done vs planned

Done now	Planned next
Login / register + users file	Full settings editor
Shell: help, ls, status, neofetch	Golf, Snake, Ludo
Candy graphics + terminal	Two-player, map tools
Ratings + average	Leaderboard command
make project	Optional terminal-only build